

Security concepts important to computer systems

Confidentiality. The information must be accessible only to the authorised people

Integrity. Guarantee that the data is what we expect

Authenticity. Guarantee that only authorised persons can access the resources

Reliability. Systems/computers should work without having unexpected problems

Availability. Services not denied to authorised users

Boot-sector viruses. Boot-sector viruses install themselves into the boot sector on a floppy disk or the master boot record on a hard disk by overwriting the original boot code with their own code. Boot-sector viruses are especially dangerous because these are executed when the user boots the computer from the disk. A boot-sector virus in the master boot record is very harmful because on each start-up of the computer, it is loaded into the memory, from where it can spread to other parts of the hard disk and result in complete system crash. In such cases, the user's computer becomes unable to start-up or even find its hard drive.

Macro viruses. Macro viruses are written in the macro/scripting languages provided with many applications such as MS Office. These viruses spread easily because they travel in documents and spreadsheets. They can spread with infected file sharing from one computer running on an operating system (Windows) to another computer running on a different operating system (Macintosh). Most of these macro viruses have the ability to replicate themselves by sending infected e-mails to everyone they find in the user's contacts.

Stealth viruses. Stealth viruses intercept read requests to the operating system. These viruses make the operating system unaware by modifying and forging the results of calls to functions in the infected file, so the system believes it is reading the original file. Such interception is obtained by malicious code injection of the actual operating system files that would handle the read request. This will result in either denial of the read request or serve the read request with an uninfected version of the file.

These viruses can also trick an antivirus software by intercepting its

read request to the operating system, handling the request itself, and returning an uninfected version of the file to the antivirus software. In this way, stealth viruses can sometimes fool the antivirus software into concluding that the system is free from viruses. Such viruses even go to great lengths to hide the fact that these are consuming memory.

Self-modifying viruses. Generally, most of the antivirus software search, in the files to be scanned, for virus signature that is nothing but a sequence of some bytes or a string. Upon detection of such patterns or strings, the antivirus software reports that the file is infected with virus. Self-modifying viruses are cleverly designed to modify their sequence of bytes on each infection. Thus detection of such viruses becomes difficult for antivirus software programs that rely on virus signatures only.

Encrypted viruses. Encrypted viruses make use of a variable key to encipher their code and also consist of a small decrypting module. Due to the use of variable key for encryption of the virus, it becomes difficult for the antivirus software to detect such viruses. The only thing that may be suspicious to antivirus software is the decrypting module.

Polymorphic viruses. Similar to encrypted viruses, polymorphic viruses use an encrypted copy of themselves. In addition, polymorphic viruses modify decryption module on each infection. Such viruses pose serious problems to antivirus software by changing their signature every time they infect a new host. Thus it becomes very difficult to detect such viruses using signatures. However, these viruses can be detected using statistical pattern analysis.

Metamorphic viruses. A metamor-

phic virus attempts to defeat antivirus software by rewriting itself completely each time it infects a new executable file. For this purpose, a metamorphic engine is used that consists of large bytes of code.

Computer worms

A computer worm is a self-replicating standalone program that independently spreads without attaching itself to an existing program. Worms often use a computer network to spread themselves very rapidly and cause a lot of damage (Fig. 2). Some worms only spread themselves without causing any harm to the system they pass through and thereby consume bandwidth up to a large extent.

Some worms, depending on the type of payload carried by them, can encrypt files or send important information via e-mail. The worm scans the infected computer for files such as address books and temporary webpages that contain e-mail addresses. It then uses the addresses to send an infected e-mail, often spoofing the address line in subsequent e-mail messages so that those infected messages appear to be from someone known.

Many worms are used to install backdoor programs on the targeted machines to take control of these machines for sending malicious mails or spam. In this way, attackers hide themselves under the group of such networked machines called 'botnets.'

Computer worms can also be used for good purposes, such as to fix vulnerabilities in the host system while exploiting the same vulnerability to download and install patches from the legitimate software manufacturer. Many worms take advantage of a vulnerability in the operating system to spread. If the vulnerability is disclosed and patched by the vendor before attack, a zero-day cyber attack is possible.

Trojan horse

Trojan horse, or Trojan, is a malicious software program that does not replicate itself and is technically not a virus. It is spread by viruses, worms